

1. Let $a_n = n^{\frac{1}{n}} : n \geq 2$ and $f(x) = \frac{\sin(x^2 - 1)}{(x - 1)}$ for $x \neq 1$. Find $\lim_{n \rightarrow \infty} f(a_n)$.

$$a_n = n^{\frac{1}{n}} \Rightarrow \ln a_n = \frac{\ln n}{n}$$

Q1 ok

So first, we should find $\lim_{n \rightarrow \infty} \frac{\ln n}{n}$

Q2a good

$$\text{Since } \lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0, \quad \lim_{n \rightarrow \infty} a_n = 1$$

Q3 (b) -10

Then we should find $\lim_{x \rightarrow 1} f(x)$

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{\sin(x^2 - 1)}{x - 1} = \lim_{x \rightarrow 1} \frac{\sin[(x-1)(x+1)]}{x-1}$$

Since $x \rightarrow 1$, $x \rightarrow 1 \rightarrow 0$

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{\sin[(x-1)(x+1)]}{x-1} &= \frac{\lim_{x \rightarrow 1} \{\sin[(x-1)(x+1)]\}'}{\lim_{x \rightarrow 1} (x-1)'} \\ &= 2 \quad \checkmark \end{aligned}$$

2. Decide if each of the following statements are true or false. In the true case, justify; in the false case, give a counterexample.

a) If $\lim_{n \rightarrow \infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ is convergent.

b) $\sum_{n=1}^{\infty} \frac{2^n + 4^n}{\pi^n}$ is divergent.

c) $\sum_{n=1}^{\infty} \sqrt[3]{n \sqrt{\frac{2}{n^4}}}$ is divergent.

a) False.

b) ① true

For example, $\sum_{n=1}^{\infty} \frac{1}{n}$

② true

is not convergent

3. Analyze the convergence of the series

$$a) \sum_{n=2}^{\infty} \frac{1}{n \ln(n)},$$

$$b) \sum_{n=2}^{\infty} \frac{1}{n [\ln(n)]^n}.$$

a) divergent

Prove:

We can consider $\int_2^{\infty} \frac{1}{x \ln(x)} dx$

assume $u = \ln(x)$, $du = \frac{1}{x} dx$

$x=2$, $u = \ln 2$, $x \rightarrow \infty$, $u \rightarrow \infty$

$$\int_2^{\infty} \frac{1}{x \ln x} dx = \int_{\ln 2}^{\infty} \frac{1}{u} du$$

We can easily find it is divergent

In conclusion, $\sum_{n=2}^{\infty} \frac{1}{n \ln(n)}$ is divergent

b) convergent

3.1

4. a) Approximate the series $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ by using the sum of the first 10 terms. Estimate the error involved in this approximation.
- b) How many terms are required to ensure that the partial sum S_N of the series

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$$

is exact to within 5×10^{-4} ?

a) We should find $\sum_{n=2}^{11} \frac{1}{n(\ln n)^2}$ first

$$\sum_{n=2}^{11} \frac{1}{n(\ln n)^2} \approx 1.7004$$

then we should find $\int_2^{\infty} \frac{1}{x(\ln x)^2} dx$

assume $u = \ln x$, $du = \frac{1}{x} dx$

when $x=2$, $u = \ln 2$

$$\int_2^{\infty} \frac{1}{x(\ln x)^2} = \int_{\ln 2}^{\infty} \frac{1}{u^2} du$$

$$\int_{\ln 2}^{\infty} \frac{1}{u^2} du = 1.4427$$

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3.1 use root test
 $L=0<1$